Benchmarking Foundation Models using Satellite Image Time Series for Land Monitoring - MVA 2025/2026

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April 3, 2026

1 Introduction

2 Dataset

We use the dataset PASTIS (Panoptic Agricultural Satellite TIme Series) There are 2433 sequences of multi-spectral of shape 10*128*128.Each sequences contains between 38 and 61 images. The satellite THEIA provides observations. The 10 channels correspond to the non-atmospheric spectral bands of the Sentinel-2 satellite, after atmospheric correction and re-sampling at a spatial resolution of 10 meters per pixel. Observatios are taken in several region of France with different climates, (Normandie, Bourgogne,Alsace, Provence). Almost 30 % of pictures are covered by cloud. **[dumeur]**

Each pixel is associated with a semantic label of 18 crop labels

3 Methods

3.1 TESSERA

TESSERA [1] (Temporal Embeddings of Surface Spectra for Earth Representation and Analysis) operates under the "Embedding-as-Data" paradigm, providing analysis-ready, pre-computed embeddings for each pixel at a global scale. The model exclusively processes time series at the pixel level and does not utilize explicit spatial features or spatial convolutions. Consequently, it achieves a direct multi-temporal fusion of the data, relying purely on the natural spatial correlation of real-world observations to form spatially coherent outputs.

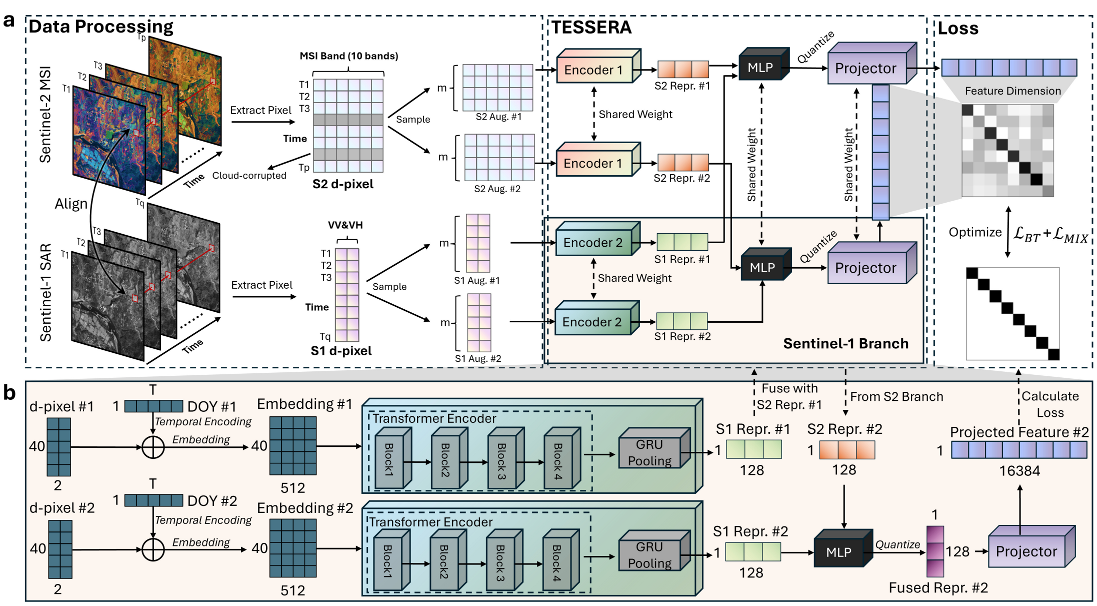


Fig. 1 (a) Overall architecture: multi-temporal Sentinel-1/2 observations are converted into modality-specific d-pixels, augmented twice, and encoded by dual branches with shared weights to produce compact 128-D embeddings. (b) Zoom-in on the Sentinel-1 branch: temporal sequences are processed by a 4-block Transformer followed by GRU pooling to capture dynamic backscatter patterns before fusion into the joint TESSERA embedding. Courtesy of [1].

As illustrated in the overview of the TESSERA processing pipeline (Fig. 1), the architecture handles multi-modal data streams by extracting temporal sequences of optical (Sentinel-2) and radar (Sentinel-1) observations into modality-specific "d-pixels". The training strategy is akin to contrastive learning, specifically employing the Barlow Twins objective function. The model aims to maximize the similarity between two different "views" of the same geographic location. In this framework, a view corresponds simply to a time series of multiple modalities randomly sampled from the available observations within the same year. This sparse temporal sampling strategy forces the model to learn representations that remain robust and invariant to the irregularity and cloud obstructions typical of Earth Observation data. It produces 128-dimensional embeddings for each pixel. This model is open, with the code, weights, and pre-computed embeddings available on the web.

3.2 AlphaEarth Foundations

In contrast to TESSERA's pixel-wise and primarily dual-modality focus, AlphaEarth Foundations [2] integrates a broader array of sensors and modalities. Its input sources and reconstructed outputs go far beyond standard optical and radar satellite imagery (Sentinel-1, Sentinel-2, Landsat 8/9), incorporating digital elevation models, LiDAR profiles (GEDI), and even continuous climate variables (ERA5-Land). By fusing these irregular and unaligned sources, AlphaEarth creates a different embedding space, representing the Earth's surface as a regular 64-dimensional embedding field.

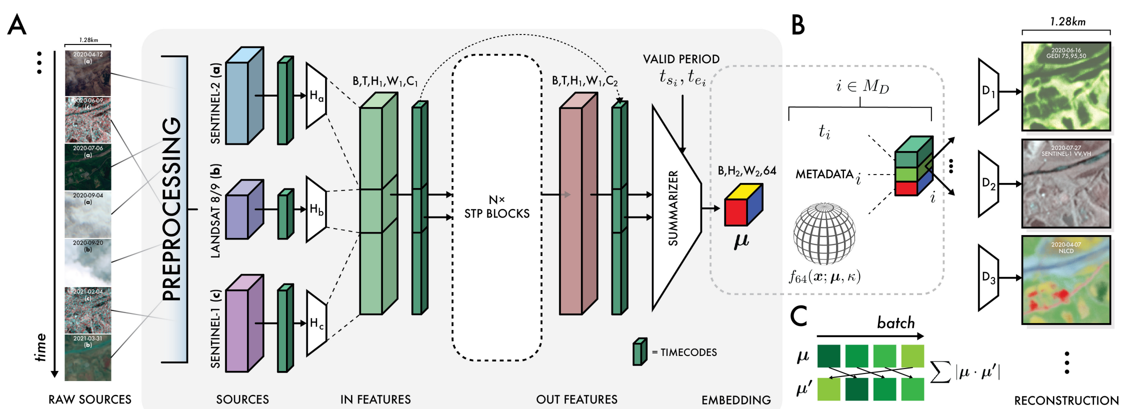


Fig. 2 (A) Block diagram of the overall network architecture used for video analysis. Preprocessing converts raw observation data via normalization using global statistics, and acquisition timestamps are converted to sinusoidal timecodes. Individual source encoders transform inputs to the same latent space before entering the bulk of the model. Outputs are summarized using conditional timecodes or "summary periods", unique to each decoded source and contrastive learning task. μ refers to the embedding outputs of the model. The grey shaded region constitutes the model. (B) Model outputs are treated as the mean direction of a von Mises-Fisher distribution, and decoding proceeds by sampling this distribution, and concatenating it with sensor geometry metadata and a timecode indicating the relative position in the valid period to decode. Decoding proceeds for all sources, with losses dependent on the characteristics of each source (see supplemental materials S1). Courtesy of [2].

The block diagram in Figure 2 depicts how the architecture summarizes these diverse spatial and temporal contexts into a highly compact 64-byte representation. Furthermore, its learning paradigm differs significantly; rather than a contrastive approach, AlphaEarth relies on a generative learning approach. The network employs an implicit decoder that attempts to reconstruct the original, multi-source observations and metadata directly from this small embedding. By working at the patch level rather than the pixel level and focusing on reconstruction, this generative approach allows AlphaEarth to explicitly capture spatial coherence and physical measurement contexts during the learning phase.

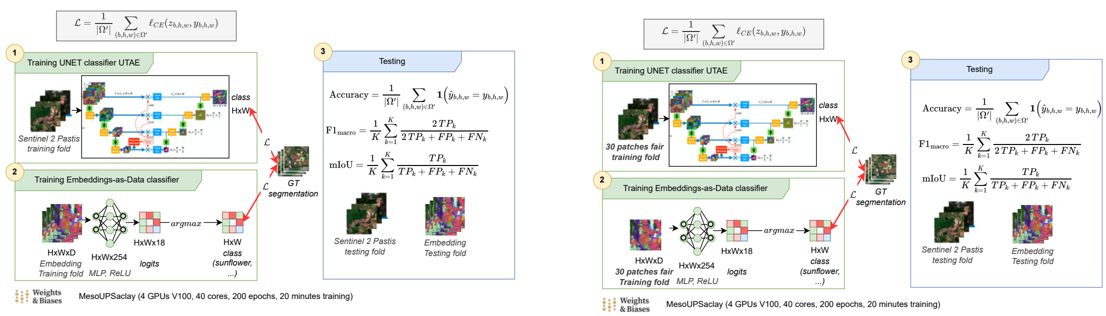
In contrast to TESSERA, AlphaEarth's code and weights are not publicly available, and the embeddings are only available under a commercial API.

3.3 Models overviews

4 Experiments and results

4.1 Benchmarking setup

Our main benchmarking task is to predict the crop class of each pixel over the PASTIS dataset ("semantic segmentation").



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(a) With the full PASTIS dataset

(b) With scarce data (30 patches per fold)

Fig. 3 Training overview of our benchmarking framework. Fig. 3a shows the training setup with the full PASTIS dataset, while Fig. 3b shows the training setup with scarce data (30 patches per fold).

Our project mentor gave us the embeddings of the three foundation models (TESSERA, Alpha Earth and ALISE) on the PASTIS dataset. Producing those embeddings is computationally expensive (or even cost money, e.g. for Alpha Earth), this significantly eased our workload. Those embeddings are tensors of shape (PARCEL_ID, H, W, D) where D is the dimension of the embedding. It means that we have one embedding per pixel, which is a vector of dimension D.

As seen in Fig. 3a, we train simple MLPs with `ReLU` activations on top of the FM embeddings (one per FM embedding set, e.g one MLP for TESSERA, one MLP for Alpha Earth, one for ALISE). Those MLPs are optimized using a cross-entropy loss, which is the standard loss for multi-class classification problems. The output of the MLPs is a vector of dimension 18, which corresponds to the 18 crop labels logits. We apply a `argmax` function to the output of the MLP to obtain the final class. Note that the complexity of the classifier does not matter, as we want to evaluate the quality of the embeddings produced by the foundation models, not the capacity of the classifier.

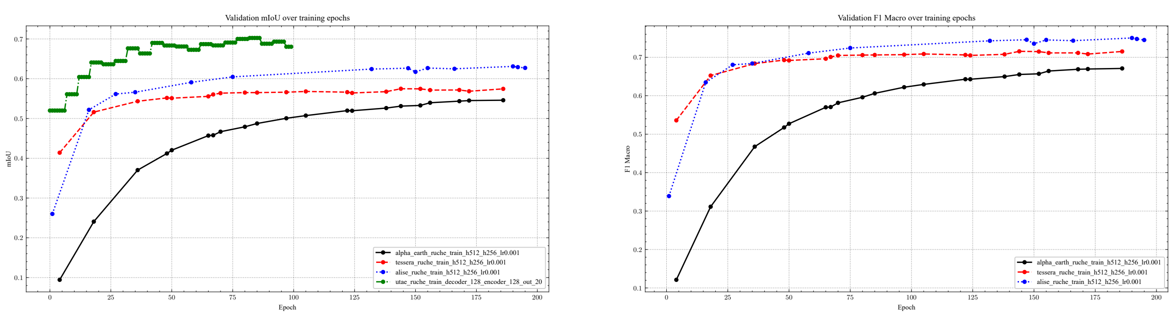
We use the same folds as the PASTIS paper, 3 for training, 1 for validation and 1 for testing. We train the MLPs for 200 epochs with a batch size of 32 (16 for ALISE due to heavier embeddings) and a learning rate of 10^{-3} with Adam optimizer. We used the Ruche cluster of Université Paris Saclay. Specifically the slurm partition `GPU-test` of Ruche cluster, with a training of roughly 20 minutes with four V100 GPUs using `DistributedDataParallel` strategy of [3]. The code is implemented using [4] for industry-grade training and logging, and monitored using [5]. All our training configurations are available in the code repository. We also provide a reproducible notebook to plot all the figures of this report.

To compare with a supervised baseline, we also train a U-TAE model on the same task. We reuse the code of the original paper for training and evaluating. We also use the same hyperparameters. Training takes roughly 14 hours on one V100 GPU. Instead of FM embeddings, it is directly trained on the optical channels of the Sentinel-2 satellite.

We evaluate the performance of the models using three metrics: mean Intersection over Union (mIoU), F1 Macro and Overall accuracy. Note that the `background` and `void` classes of the PASTIS dataset are ignored in the evaluation (as well in the training loss of the MLPs and U-TAE).

We also evaluate the performance of the models in a scarce data setting, where we only use 30 patches per fold for training (configurations suffixed with "_scarce"), see Fig. 3b. Those folds are randomly sampled from the original training folds but maintain a balanced class distribution. This scarce data setting is particularly relevant for satellite image segmentation, as it is often the case that we have limited labeled data.

4.2 Analysis of results



(a) **(b)**

Fig. 4 Increasing trends of the validation F1 Macro (Fig. 4b) and mIoU (Fig. 4a) metrics during training (validation fold). As shows Fig. 4a, U-TAE is trained only for 100 epochs. Fig. 4b does not show the U-TAE curve as the original implementation does not log the F1 Macro metric during training.

Over training epochs, benchmarked models successfully learn to segment the images, as shown by the increasing trends of the validation F1 Macro and mIoU metrics in Fig. 4.

As shown in Table 1, the supervised approach U-TAE achieves the best performance on the full training set, while MLP+ALISE-EMBEDDINGS performs best in the scarce data setting. Despite Tessera (released in November 2025) and ALISE (released in September 2024) being separated by one year of research, ALISE is still a State-Of-The-Art FM in our benchmark. The scarce data setting shows that foundation model achieve strong performance when we have limited labeled data, as they can leverage their pre-training on large amounts of unlabeled data to produce useful embeddings for downstream tasks. Note that there is a net computational cost to using foundation models, as the training of the MLPs on top of the FM embeddings is much faster than training a supervised model like U-TAE from scratch, with less parameters.

As noted in [7, 6], the PASTIS dataset is highly imbalanced, with a ratio of over 50 between the

Table 1 Test metrics of the FM-EMBEDDINGS+MLP / U-TAE models on the PASTIS dataset, with the full training set (Table 1a) and with scarce data (Table 1b). Bold values indicate the best performance for each metric.

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.5287	0.6526	0.8417
Tessera	0.5607	0.7018	0.8257
ALISE	0.6009	**0.7235**	0.8651
U-TAE	**0.7219**	-	**0.9170**

(a) Test metrics

Model	mIoU	F1 Macro	Overall accuracy
AlphaEarth	0.2867	0.3755	0.7267
Tessera	0.4387	**0.5765**	0.7621
ALISE	**0.4482**	0.5729	**0.7898**
U-TAE	0.3869	0.5115	0.7043

(b) Test metrics with scarce data (30 patches per fold)

Table 2 Test F1 per class

	test_alpha_earth_ruche_h512_h256_lr0.001	test_tessera_ruche_h512_h256_lr0.001	test_alise_ruche_h512_h256_lr0.001
Meadow	**0.9217**	**0.8859**	0.9216
Soft winter wheat	0.8793	0.8501	0.9155
Corn	0.9184	0.8572	0.9306
Winter barley	0.6718	0.7947	0.8354
Winter rapeseed	0.9124	0.8734	**0.9361**
Spring barley	0.5308	0.6404	0.7050
Sunflower	0.6303	0.7465	0.6896
Grapevine	0.8745	0.8561	0.8803
Beet	0.8953	0.8247	0.9289
Winter triticale	0.3222	0.4492	0.4334
Winter durum wheat	0.7250	0.7168	0.7979
Fruits, vegetables, flowers	0.4885	0.5821	0.5547
Potatoes	0.6144	0.5599	0.6988
Leguminous fodder	0.4339	0.5538	0.4998
Soybeans	0.8129	0.8238	0.8646
Orchard	0.7246	0.6713	0.6528
Mixed cereal	0.1472	0.4740	0.3414
Sorghum	0.2433	0.4725	0.4367

most and least common classes (see Fig. 5). Thus, we provide in Table 2 the F1 score of each class for the different models, which gives a more detailed view of the performance of the models on each class.

We also show on Fig. 6 the confusion matrix of each FM-EMBEDDINGS+MLP model. Unsurprisingly, confusions seem to occur between semantically close classes such as different cereal types, or Sunflower and Fruits, Vegetable, Flower.

Finally, we show in Fig. 7 and Fig. 8 the predicted semantic segmentation maps on a test patch. The scattered pixels on Tessera predictions shows that it work at a pixel level, while the smoother predictions of ALISE and Alpha Earth show that they work at the patch level. Note that in those prediction maps, the background and void classes are masked out (transparent pixels), which lead to clearly defined fields boundaries. While this is convenient for visualization, it is not the case for industry applications, where we often need to predict the background and void classes as well. Thus, it would be interesting to evaluate the performance of the models on those classes as well, which is left for future work. In the scarce data setting, we can see that the predictions are much noisier for U-TAE, which is consistent with the performance drop of U-TAE in the scarce data setting compared to the FM-EMBEDDINGS+MLP models.

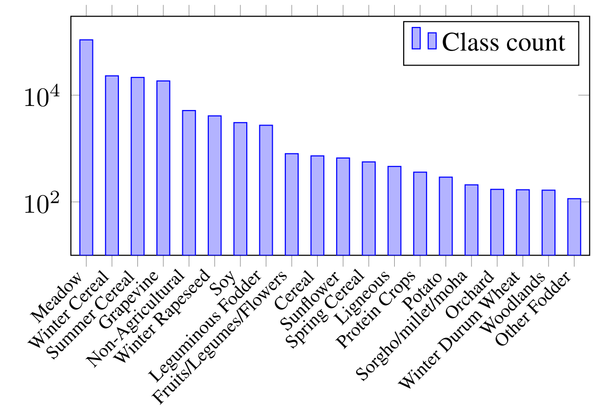
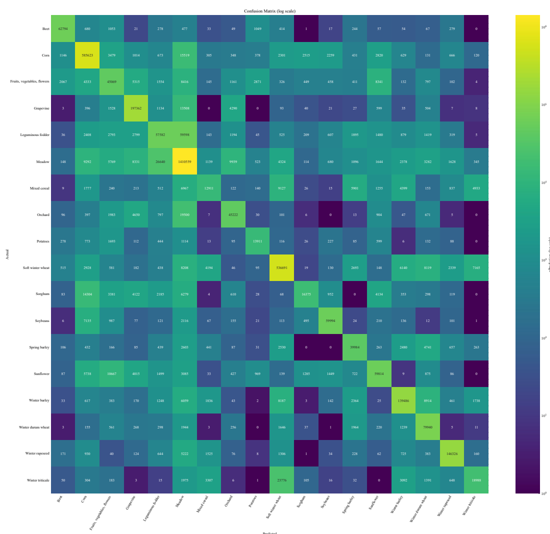
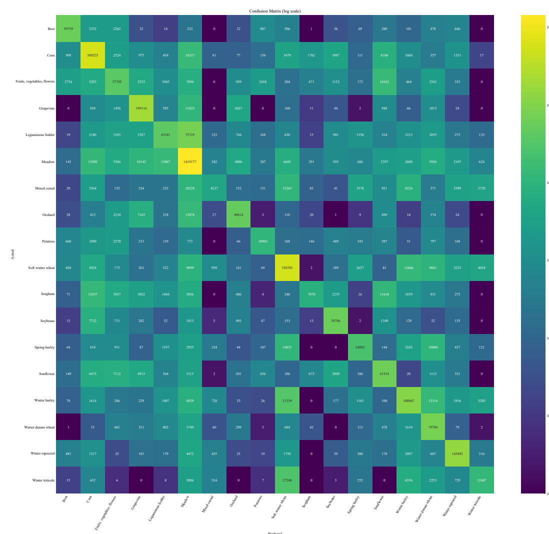


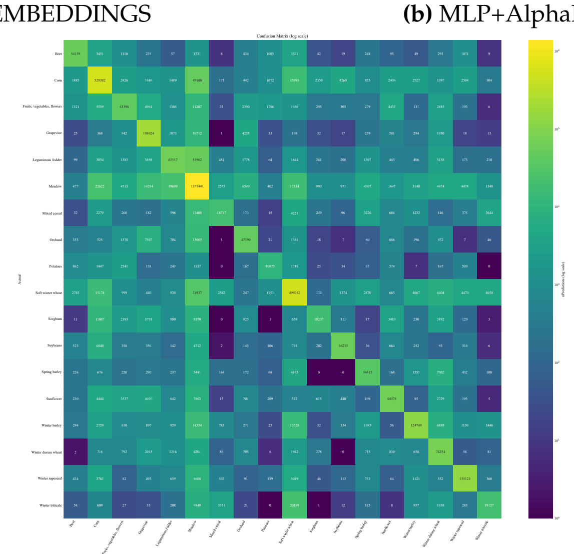
Fig. 5 Class distribution of the PASTIS dataset. Courtesy of [6]



(a) MLP+ALISE-EMBEDDINGS



(b) MLP+AlphaEarth-EMBEDDINGS



(c) MLP+Tessera-EMBEDDINGS

Fig. 6 Confusion matrices on the test fold for foundation models.

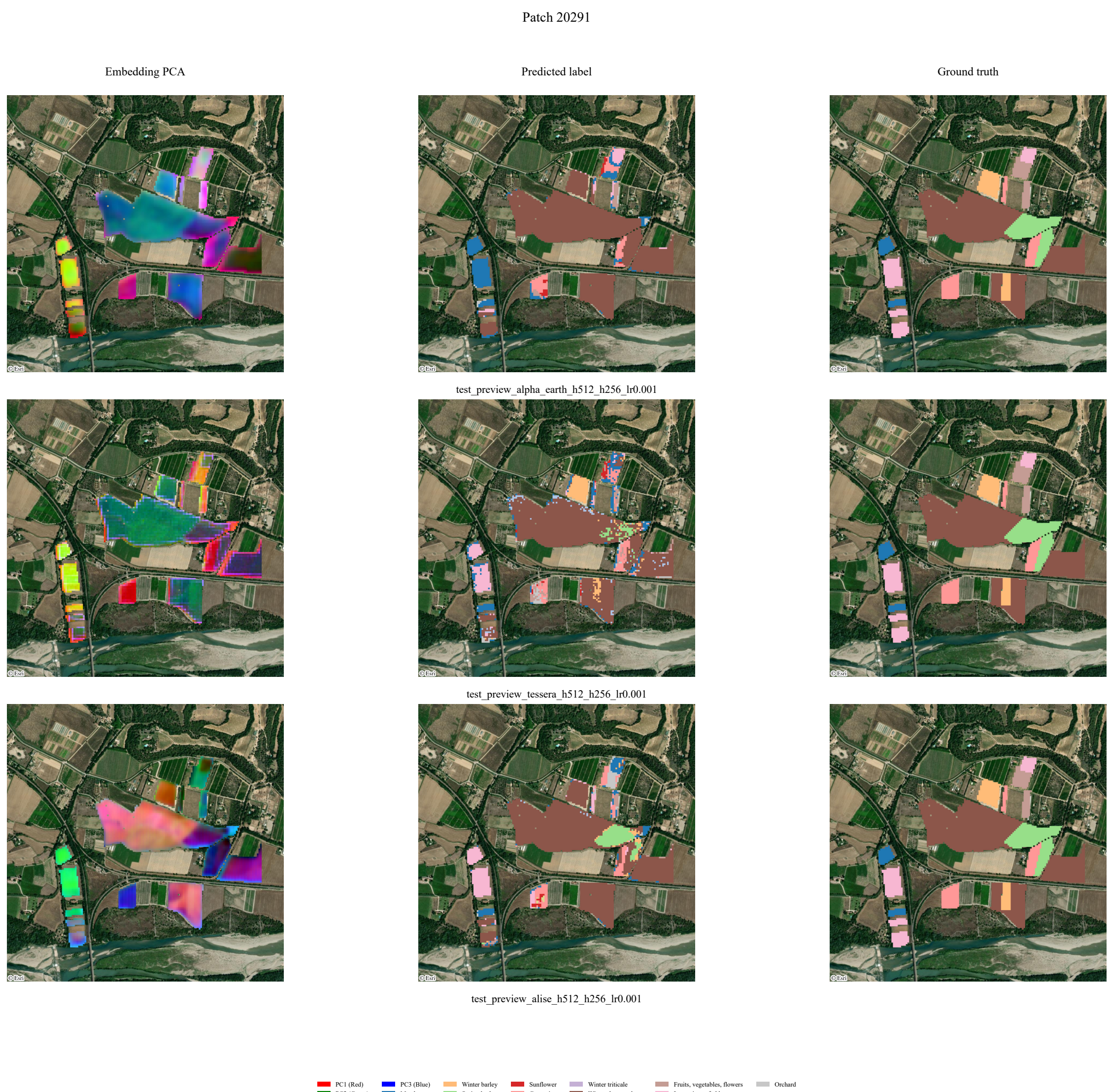


Fig. 7 Semantic segmentation map (full dataset). Row 1 : Alpha Earth, Row 2 : Tessera, Row 3 : ALISE. The first column shows the first three PCA components of the embedding, the second column shows the predicted segmentation map, the third column shows the ground truth segmentation map.

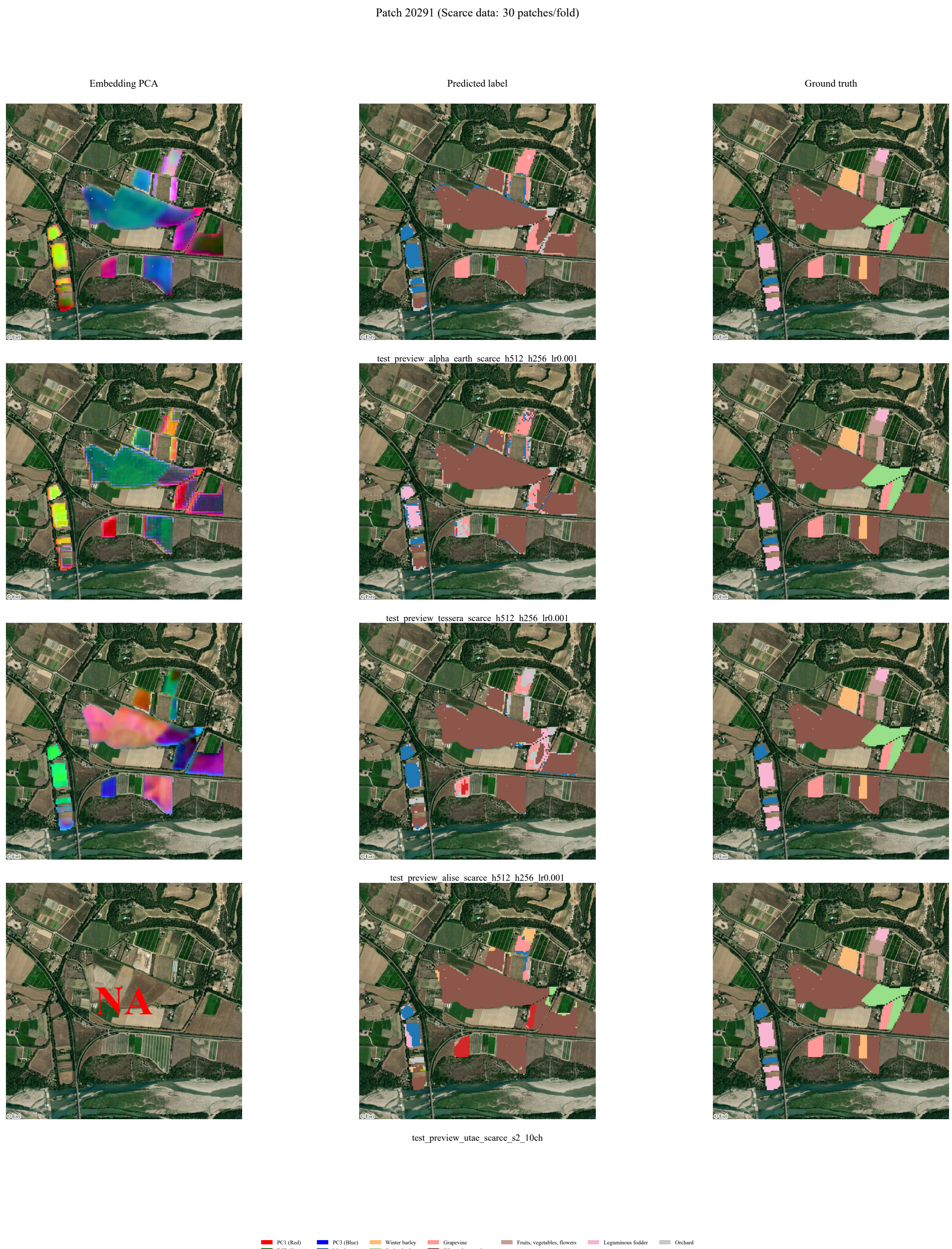


Fig. 8 Semantic segmentation map (scarce dataset). Row 1 : Alpha Earth, Row 2 : Tessera, Row 3 : ALISE, Row 4 : U-TAE. The first column shows the first three PCA components of the embedding, the second column shows the predicted segmentation map, the third column shows the ground truth segmentation map. Note that U-TAE embeddings are not available, as U-TAE is trained directly on the optical channels of the Sentinel-2 satellite.

5 Summary of contributions and future work

In this report, we propose a standard benchmarking framework to evaluate the performance of foundation models on the task of satellite image segmentation, reproducible and maintainable over time. We evaluate three state-of-the-art foundation models (TESSERA, Alpha Earth and ALISE) on the PASTIS dataset, and compare their performance to a supervised baseline (U-TAE). We show that while supervised methods achieve the best performance on the full training set, FM performs best in the scarce data setting.

Future work could include evaluating the performance of the models on other tasks, such as predicting the background and void classes. It would also be interesting to try other geographical regions, as the PASTIS dataset is limited to France, or even evaluate the transferability of the trained models to different regions on the same task. It would be interesting to include more foundation models in the benchmark as new models are released. We could further improve this benchmark by proving a website with an interactive interface to visualize the results and compare the different models, with a SDK to easily add new models to the benchmark, like https://timeseriesclassification.com/. The practicability of foundation models (e.g. easy training, whole Earth coverage, etc.) is good for prototyping applications, such as a crop monitoring application e.g, "what-is-this-field.net"

References

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This work was performed using computational resources from the "Mésocentre" computing center of Université Paris-Saclay, CentraleSupélec and École Normale Supérieure Paris-Saclay supported by CNRS and Région Île-de-France (https://mesocentre.universite-paris-saclay.fr/).

Appendix A Method

Appendix B Data

Appendix C Tables